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9702/32

May/June 2010

2 hours

Additional Materials: As listed in the Confidential Instructions.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

Additional answer paper and graph paper should be submitted only if it becomes necessary to do so. You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
All questions in this paper carry equal marks.

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2	
Total	

This document consists of **9** printed pages and **3** blank pages.

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1 In this experiment, you will measure the current through a set of resistors.

- (a) (i)** Connect the circuit of Fig. 1.1, ensuring that the movable lead is connected between resistors 1 and 2.

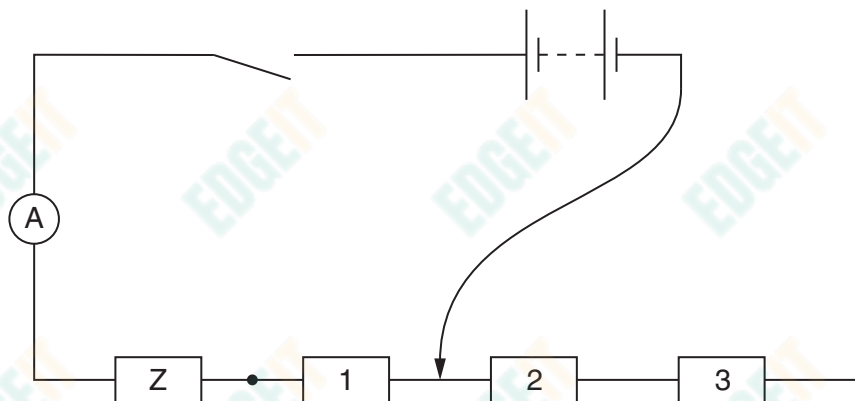


Fig. 1.1

- (ii)** Close the switch and record the ammeter reading I . Open the switch after recording your measurement.

$I = \dots\dots\dots$ A

- (b) By adjusting the movable lead, the resistor Z may be connected in series with a number N of other resistors. Each of the resistors labelled 1 to 12 has a resistance of $22\ \Omega$.

Repeat (a)(ii) for different values of N until you have six sets of readings for N and I .

Include in your table of results values of $\frac{1}{I}$ and the total resistance R of the $22\ \Omega$ resistors connected into the circuit.

(The total resistance R of the $22\ \Omega$ resistors in series can be determined using the formula $R = R_1 + R_2 + R_3 + \dots$)

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- (c) (i) Plot a graph of R on the y-axis against $\frac{1}{I}$ on the x-axis.
(ii) Draw the line of best fit.
(iii) Determine the gradient and y-intercept of the line of best fit.

gradient =

y-intercept =

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- (d) The quantities R and I are related by the equation

$$R = \frac{G}{I} + H$$

where G and H are constants.

Use your answers to (c)(iii) to determine values for G and H . You should include units where appropriate.

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$G =$

$H =$

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- 2 In this question, you will investigate how the attractive force between two magnets depends on their separation.
- (a) You are provided with a pair of magnets, one of which is fixed to the bench. You are also provided with some microscope slides, adhesive tape, a loop of cotton and a newton meter. You also have access to a micrometer screw gauge.
- (i) Place the loop of cotton around the magnet that is **not** fixed, as shown in Fig. 2.1, and use the adhesive tape to secure the cotton to the sides.

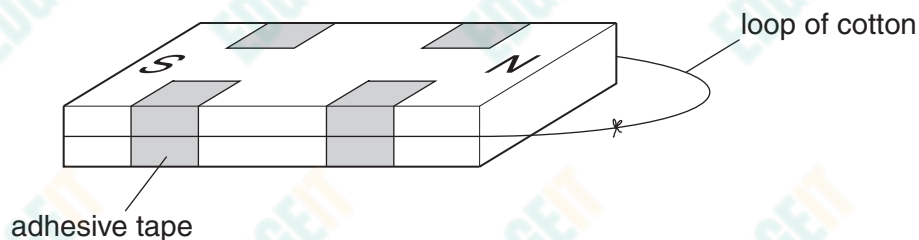


Fig. 2.1

- (ii) Place the magnet with the loop of cotton end-to-end with the fixed magnet, so that they are attracting, as shown in Fig. 2.2.

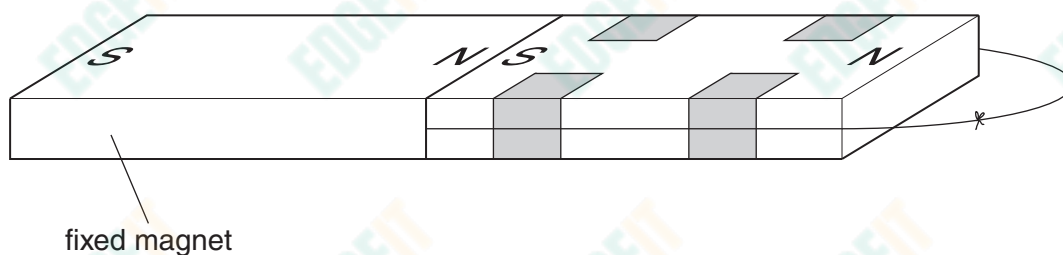


Fig. 2.2

- (b) (i) Using the newton meter, measure the maximum force F required to pull the magnets apart, as shown in Fig. 2.3.

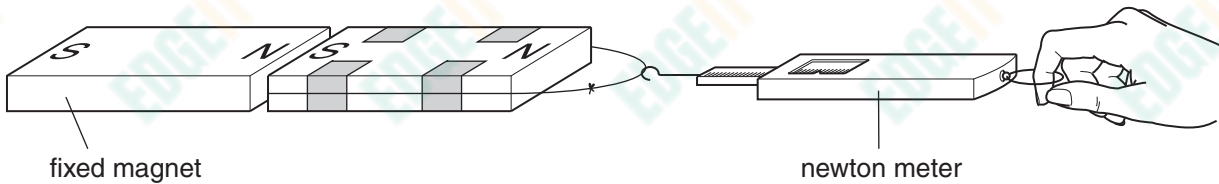


Fig. 2.3

$F = \dots\dots\dots$ N

- (ii) Explain why it is difficult to determine this force accurately.

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- (iii) Estimate the percentage uncertainty in the maximum force required to separate the attracting magnets. Show all your working.

percentage uncertainty =

- (c) (i) Using the micrometer screw gauge, measure the total thickness t of three of the microscope slides, as shown in Fig. 2.4.

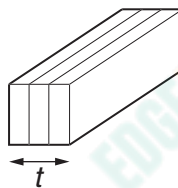


Fig. 2.4

$t = \dots\dots\dots$ mm

- (ii) Explain how you have made this measurement as accurate as possible.

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- (iii) Place the three slides between the two attracting magnets, as shown in Fig. 2.5. Use the newton meter to find the maximum force F needed to separate the magnets.

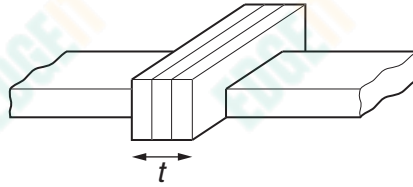


Fig. 2.5

$F =$

- (d) Repeat (c)(i) and (c)(iii) for a single slide placed between the attracting magnets.

$t =$ mm

$F =$

- (e) It is suggested that F and t are related by the equation

$$F = \frac{k}{t}$$

where k is a constant.

By calculating values of k , explain whether your results support this relationship.

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- (f) (i) Describe four sources of uncertainty or limitations of the procedure in this experiment.

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- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

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